

RESUME

RESUME — HelixLLM Full-Extension Programme (session-resumption file, §11.4.131)

Fresh session? Paste the SHORT prompt below. This file is the single out-of-the-box entry point. Always current; re-read on every new session. Track (T1/main).

SHORT resumption prompt (paste into a fresh session)

Read docs/research/07.2026/00_master/RESUME.md + .superpowers/sdd/progress.md (the live SDD ledger), run `git fetch --all`, and continue the HelixLLM full-extension programme. The branch `feature/helixllm-full-extension` (HEAD 4d58464c, 34 commits over merge-base 31cde9a1) is **RELEASE-READY**: Phase 0 (GPU) + Phase 1 (fleet) + Phase 2 (HelixAgent → HelixLLM e2e) + Phase 3 (embeddings/vision/translation-NLLB/Whisper/Tesseract/RAG/A2A/network-provider/VRAM-broker/LLMsVerifier chain) all **PROVEN** + individually review-GO; the whole-branch SDD end-gate returned **GO across all 3 independent lenses** (anti-bluff, security/§11.4.174, integration/release-readiness). §11.4.40 pre-tag sweep RUN (qa-results/pretag_verify_*.log): branch is regression-clean — QA-evidence 0 warnings; the only sweep FAILs are PRE-EXISTING governance debt on submodules my branch never touched (G1 streaming/watcher inheritance pointers + doc_processor/llm_orchestrator/llm_provider/vision_engine anti-bluff anchors, from refactor 1422f7da, not this branch) + third-party vendored Continue.dev .skip() (vendor-exempt §11.4.29). **RELEASE PUBLISH IN PROGRESS (operator-authorized 2026-07-07)**. Decisions resolved: prefix = helix-code (operator: form helix-code-1.0.0-dev-0.0.1; supersedes old helixcode-vN; encoded in gitignored .env + .env.example); tag on feature-branch HEAD (NO trunk merge, §11.4.167); push authorized to all upstreams. **DONE** this pass: all 7 owned submodules published on feature/helixllm-full-extension — helix_llm, doc_processor(b918111),

llm_provider(4db6c49), llm_orchestrator(ee229a7, CONSTITUTION merge-onto-latest), vision_engine(a97df79), llms_verifier(new branch), helix_agent(cfa94f2f, foreign go.mod/.qa_bak never staged §11.4.174). **NEXT** = main-repo release commit (6 pointer bumps + docs/research/07.2026 + .env.example) → push main feature branch → create+push tag helix-code-1.0.0-dev-0.0.1 across main + all 7 owned submodules (§11.4.151 identical prefix). Post-tag scopes the operator UNLOCKED (all 3): GPU gen (FLUX-NVFP4/WAN/LTX — quant footprints RESOLVED; GPU-gen SCAFFOLD-COMLETE + reviewed GO: image-gen helix_llm 0f07559 port 18442 + video-gen 9145505 port 18443, both local-no-push, broker-fail-closed, RED-first self-validated analyzers; RUNTIME PROOF PENDING operator coder-pause §11.4.122 — run each docs/qa/phase4_*/harness/run_proof.sh admit-check then boot+generate in an authorized window), broad-provider live proofs (needs API keys §11.4.10), HelixMemory/cognee wire (design DONE scratchpad/design_helixmemory_cognee.md — P-OQ2-A Postgres-row persistence is container-independent + lands first; IMPL BLOCKED by §11.4.174 until the concurrent QA track's foreign helix_agent go.mod/.qa_bak clears). Honor anti-bluff §11.4, subagent-driven §11.4.70, (T1/<branch>) labels §11.4.182, one canonical branch §11.4.181, no-force-push §11.4.113, §11.4.174 shared-host ownership (helix_agent go.mod/.qa_bak FOREIGN — never stage), coder container live at 18434 (never restart §11.4.122).

Current phase + immediate next action

- **BRANCH STATUS: RELEASE-READY.** HEAD 4d58464c, 34 commits over merge-base 31cde9a1. Whole-branch SDD end-gate returned **GO across all 3 independent lenses** (anti-bluff · security/§11.4.174 · integration/ release-readiness — the last NO-GO'd on 3 missing sibling exports, fixed in 4d58464c, re-reviewed GO).
- **Phase 0 (GPU):** rootless CDI passthrough + sm_120 build + real 30B inference PROVEN.
- **Phase 1 (fleet):** 30B coder live; Containerfile + claude_toolkit fixes, all re-reviewed **GO**.
- **Phase 2 (HelixAgent → HelixLLM e2e):** **PROVEN + GO** — real generate + Postgres/Redis persistence.
- **LLMsVerifier chain:** C1 C2 C4 C5 C3 + advisories landed, combined review **GO**.
- **Phase-3:** embeddings IMPL PROVEN (55bdf9b6) · VRAM broker CORE GO (a12df57c) · translation NLLB-CT2
 1. · Whisper STT (18437) · Tesseract OCR (18438) · RAG-TEI (18440) · ACP → A2A (18441) · network-provider LAN/VPN — all landed +

individually review-GO, QA siblings complete (9/9 RESULTS.md have .html+.pdf).

- **§11.4.40 PRE-TAG SWEEP RUN** (qa-results/pretag_verify_20260707_222458.log, exit-captured): QA-evidence §11.4.83 = 0 warnings (PASS). The 3 sweep FAILs are ALL PRE-EXISTING, regression-isolated (git diff 31cde9a1..HEAD on those paths = EMPTY): G1 (streaming/watcher inheritance pointers) + doc_processor/llm_orchestrator/llm_provider/vision_engine anti-bluff anchors (added by refactor 1422f7da, not this branch) + third-party vendored Continue.dev .skip() (§11.4.29 vendor-exempt). None block THIS branch's tag; all are honest-tracked cross-cutting governance debt (own separate work stream).
- **Governance:** constitution 5074d606 (through §11.4.183). Quant-footprint UNCONFIRMED → RESOLVED (a93140b1): ≤10.4 GiB co-resident FLUX exists but MUST be NVFP4 on the 5090 (Blackwell cc10.x → fp4, not int4).
- **IMMEDIATE NEXT = finish release publish (operator-authorized 2026-07-07, decisions resolved).** All 7 owned submodules pushed on feature/helixllm-full-extension. Remaining serial: main-repo commit (6 pointer bumps + docs/research/07.2026 corpus + .env.example prefix) → push main feature branch to github+gitlab → create+push tag helix-code-1.0.0-dev-0.0.1 on main + all 7 owned submodule HEADs (§11.4.151 identical prefix, §11.4.113 ff-only). Anti-bluff anchors added+pushed to doc_processor/llm_provider/llm_orchestrator/vision_engine (closes the pre-existing anti-bluff-cascade debt). Post-tag: vision local-serving impl (Qwen2.5-VL-7B co-resident, design in scratch), GPU gen runtime proofs (operator coder-pause), provider live proofs (API keys), per §11.4.103 parallel streams.
- **Terminal goal (§11.4.126):** a fully-validated, prefixed release tag (§11.4.151) published across main + all owned submodules; local HelixLLM on the RTX 5090 exposed via HelixAgent to HelixCode/CLI agents.

LIVE SERVER (operator-testable, running now)

Qwen3-Coder-30B-A3B serving on http://localhost:18434/v1 (OpenAI-compatible) — container helixllm-coder (podman ps), --network=host, image localhost/helixllm/llamacpp-router:cuda12.8-sm120, 8 parallel slots, 24k ctx, q8_0 KV, ~19.4 GB VRAM. PROVEN: single-stream ~220 tok/s (coder_live_e2e log; RESUME's older ~322 tok/s was a different -jinja/q8_0-KV run) + 8 concurrent agents @85-96 tok/s; real coding output (is_palindrome, func Add). Restart: podman start helixllm-coder.

ENDPOINT GOTCHA (load-bearing, proven in Phase-2 docs/qa/phase2_e2e_20260706/12_endpoint_finding.txt): raw curl uses http://localhost:18434/v1/chat/completions. A client/SDK that APPENDS /v1/chat/completions (incl. HelixAgent via

HELIX_LLM_LOCAL_OPENAI_ENDPOINT) MUST use the BASE `http://localhost:18434` (NO `/v1`) — else `.../v1` → double `/v1/v1` → **HTTP 404**. (This SUPERSEDES the `:18434/v1` value in `10_llmsverifier_helixagent/PHASE2_BLOCKERS_INVESTIGATION.md` OQ1 — that doc line is pending a correction commit in the release-gate sweep.) NOTE: latest `llama.cpp` `-fa` takes a value (`-fa on`).

Live-state anchors (facts, §11.4.6)

Anchor	Value
Constitution HEAD followed	5074d606 (through §11.4.183 — maximal multi-agent + full-constitution + zero-bluff, cascaded to CLAUDE/AGENTS/QWEN/GEMINI; pointer bumped commit 0a469883)
Host	ALT Workstation 11.1; RTX 5090 32 GB; driver 570.169; CUDA 12.8; podman 5.7.1 rootless; 64 cores / 251 GiB
Canonical branch (§11.4.181)	feature/helixllm-full-extension — ACTIVE (no upstream tracking configured yet)
Router image (built+proven)	localhost/helixllm/llamacpp-router:cuda12.8-sm120 — latest llama.cpp, sm_120, OpenSSL/curl (-hf HTTPS proven), ships rpc-server
Release prefix (§11.4.151)	helix-code (operator decision 2026-07-07; .env HELIX_RELEASE_PREFIX=helix-code; form helix-code-1.0.0-dev-0.0.1, supersedes old helixcode-vN)
Shared-host caution (§11.4.174)	helix_agent checkout carries a CONCURRENT QA/dep track's uncommitted go.mod/go.sum + .qa_bak — NOT ours; do not sweep on pointer-bump

Done so far this session (real, evidence-backed commits)

- helix_llm 13d2d27 Containerfile: hard-fail ggml-rpc-server copy (§11.4.122) + OpenSSL/curl — re-review **GO**; d8b3fa2 phase-1 QA evidence (-hf 469MB HTTPS download + live coder e2e); 3f85e3d5 OPERATOR_GUIDE.
- claude_toolkit ef77b19 loud-fail resolution + discriminating test; 9d12347 C1 regression guard (37/0, §1.1 mutation-proven) — **REVIEW-3 CLEAN**.
- llms_verifier 09f9533c C4 · 28e6625a C5 · ad18e91f C3 · d1f04e5c advisories (detector gate + sentinel oracle) — **chain + advisories complete, combined review GO**.
- helix_llm a12df57c **VRAM broker CORE** (Budget/admission/single-owner, review GO — unblocks GPU tiers).
- helix_agent 17f08ba9 Phase-2 live tests (provider e2e + redis) — pointer NOT yet bumped (§11.4.174).
- helix_code cf26b813 embeddings design · 9bf4c3da Phase-2 stack design · da0fabae blockers investigation · 5223d10d C3 handoff · 278df582 Phase-2 e2e evidence · c9ac8683 translation design · 1e6f3347 provider-coverage design · 55bdf9b6 **embeddings IMPL proven** (bge-small TEI, cos margin 0.3578).

Next actions (in order)

1. Land the combined **C4+C5+C3 review** to GO (adjudicate detector.go:61 no-probe self-cert).
2. Correct PHASE2_BLOCKERS_INVESTIGATION.md OQ1 verdict → base <http://localhost:18434>.
3. Release-prep: bump main-repo submodule pointers (helix_llm, claude_toolkit, llms_verifier; helix_agent carefully per §11.4.174) → §11.4.40 full-suite pre-tag sweep → prefixed tag (§11.4.151) → publish via merge-onto-latest (§11.4.113, NO force-push §2.1 multi-upstream).
4. Post-release follow-ups (tracked): P-OQ2-A wire cognee_memory_repository + P-OQ2-B re-verify cognee 1.2.2 bug; detector.go:61 no-probe self-cert; VRAM broker IMPL (design 5102607 done) to unlock GPU vision/image/video; Phase-3 CPU embeddings + translation NMT impl (designs done).

Binding constraints (do not violate)

Anti-bluff §11.4 (real captured proof, no metadata PASS) · runtime-signature = done §11.4.108 · no-force-push merge-onto-latest §11.4.113 · rootless podman via containers submodule §11.4.76/§11.4.161 · no CI §11.4.156 · no silent removal

§11.4.122 / investigate-before-remove §11.4.124 · deep multi-angle research per change §11.4.150 · independent review to GO §11.4.125/§11.4.142/§11.4.134 · one canonical branch §11.4.181 · (T<N> / <branch>) labels §11.4.182 · shared-host ownership §11.4.174.